An HPSG-based Shared-Grammar for the Chinese Languages: ZHONG [|]

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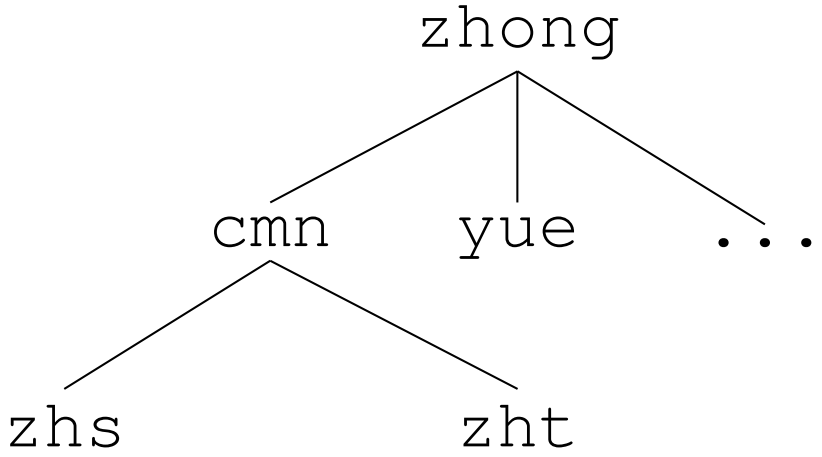
- Chinese refers to a set of various languages including Mandarin Chinese, Cantonese, Min, etc.
- These languages under the same language family share a large portion of grammatical components, though their characters and lexical items differ and some syntactic operations also differ.

Therefore: a Chinese resource capable of covering both the common parts of the grammars and the linguistic diversity across the languages.

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Structure of ZHONG [|]

- A shared-grammar for Chinese (`wiki.delph-in.net/moin/ZhongTop`)



- `zhong.tdl` is inherited by all
  - `yue.tdl` is the Cantonese grammar
  - `cmn.tdl` is inherited by the Mandarin grammars
    - ★ `zhs.tdl` is the Mandarin grammar with simplified characters
    - ★ `zht.tdl` is the Mandarin grammar with traditional characters

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Size of Grammar

items	common	cmn	zhs	zht	yue
types	383	21	1,017	7	17
phrase rules	79	0	0	0	0
lexical rules	69	4	3	0	2
lex-entry types	89	5	0	0	9
lexicon	–	–	43,067	17,470	903
testsuites	–	–	16	1	3

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Grammar Enhancement

	what we improved	what we added	what we plan to do
grammar	topic-comment clefts BA and BEI NP structures classifiers argument structure adpositions	reduplication VV compounds A-not-A questions particles fragments interjections honorification	relative clauses nominalization serial verbs conjunctions
engineering	generation	unknown word handling* bridging rules (cf. Dan Flickinger)* Generic lexical rules(regex) test modules full-forest treebanking	Wordnet incorporation transfer rules machine translation

* Robust parsing

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Evaluation (Coverage)

Coverage of simplified Mandarin (%)

	test sets	plain	unk	br	unk+br	gen
test suites	fu-berlin	22.22	80.25	22.22	97.53	90.91
	mcg-wxl	57.28	66.30	82.44	99.37	92.94
	jec	13.33	41.16	27.04	79.34	90.10
corpora	ntumc	3.47	15.58	10.54	47.82	70.54
	pctb	0.84	7.10	10.18	43.70	42.86
	sinica	3.88	40.36	6.52	65.00	80.41

Tree banking results:

- The MRS testsuite (107 sentences) - 102 parsed, 14 rejected)
- 斑点带子案 *The Adventure of the Speckled Band* (Song, 2014) (first 101 sentences) - 12 parsed, 8 rejected)

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Outlook

Where do we go from here?

- Increase the coverage of ZHONG [|] (mainly `zhs` and `yue`)
- Simultaneously build up MT (learning **and** building rules)
- Integrate with the Chinese Open Wordnet — the integrated semantic framework (ISF)

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Thanks

- The initial version of ZHONG [|] was based on ManGO (by Justin Yang and Dan Flickinger), which used the Grammar MATRIX (Bender et al., 2002) to generate the initial grammar.
- We have benefited greatly from discussions within the **NTU computational linguistics group**, as well as the wider DELPH-IN community, in particular Yi Zhang and Rui Wang
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